

# AWALUDDIN

Backend Engineer · Node.js · Go · AI Integration · Fintech & Telco Systems

Jakarta, Indonesia | +6285399589014 | awal14h@gmail.com | [linkedin.com/in/awaluddin0001](https://www.linkedin.com/in/awaluddin0001) | [github.com/awaluddin-dev](https://github.com/awaluddin-dev) | [awaluddin.dev](https://awaluddin.dev) | Open to fully remote (UTC+7)

## EXECUTIVE SUMMARY

Backend Engineer with 4+ years delivering production systems under real regulatory pressure - including solo engineering of an OJK & BI-regulated fintech platform as the sole developer under active regulatory audit. Currently consulting for Astra Group (SERA), building Node.js microservices integrated with SAP and Azure Service Bus for enterprise logistics at scale.

Proven track record: IoT infrastructure saving \$18K/year at Telkomsel, race-condition-proof financial transaction engines, and AI workflow automation using Python and LangGraph. Open to fully remote roles worldwide (UTC+7, flexible for US/EU overlap).

## TECHNICAL EXPERTISE

|                            |                                                                                              |
|----------------------------|----------------------------------------------------------------------------------------------|
| <b>Backend</b>             | Node.js (TypeScript · NestJS · Express · Inversify), Golang, Python, PHP (Laravel)           |
| <b>Architecture</b>        | Microservices, Event-Driven (Azure Service Bus),DDD, Vertical Slice, REST API                |
| <b>AI &amp; Automation</b> | LangChain, LangGraph, OpenAI API, Python async pipelines, multi-agent workflows              |
| <b>Cloud &amp; DevOps</b>  | Azure (APIM · SQL · Service Bus · Key Vault), Docker, Kubernetes, ArgoCD, Jenkins, SonarQube |
| <b>Data &amp; Storage</b>  | PostgreSQL, MongoDB, Redis, Azure SQL, MySQL                                                 |
| <b>Protocols &amp; IoT</b> | Modbus TCP, Arduino firmware, RESTful sensor APIs, OpenVPN                                   |

## PROFESSIONAL EXPERIENCE

### PT Serasi Autoraya (SERA) - Astra Group Mar 2025 - Present

#### Backend Engineer (Consultant) · Jakarta, Indonesia

- Legacy Modernization: Spearheaded migration of the Driver Management System (FMS DM) from a legacy .NET monolith to scalable Node.js microservices using Express and Inversify, secured via Azure Key Vault with SonarQube and pen-testing for code quality.
- Event-Driven Integration: Designed a decoupled Azure APIM + Service Bus integration layer with async listener services to ingest high-volume payroll and clock-in data from Mekari Talenta, SAP, and FMS 2.0 - zero data packet loss.
- DevOps & Storage: Configured CI/CD pipelines (Jenkins, GitHub, ArgoCD) with split storage paths: Azure SQL for transactional data, MongoDB for audit logs, Redis for session management.

### Telkomsel (via PT Deta Sukses Makmur) Apr 2024 - Mar 2025

#### Software Engineer (Contract) · Makassar, Indonesia

- Infrastructure Savings (\$18K/year): Eliminated vendor dependencies by assembling custom on-premise servers and deploying bare-metal Kubernetes on Ubuntu, saving \$18,000 annually in licensing costs.
- IoT Backend & Hardware Integration: Built a Golang data ingestion service polling Modbus power meters and Arduino sensors (temperature, humidity, fuel) every 5 minutes - serving a real-time Vue.js dashboard and Flutter mobile app to 150+ internal users.
- System Impact: Achieved 99.9% uptime and cut incident response time from hours to under 5 minutes, replacing 3x daily manual checks with automated monitoring.

### Regulated Fintech Company (OJK & BI Regulated) Apr 2023 - Mar 2024

#### Full Stack Developer · Makassar, Indonesia

- Compliance Platform Modernization: Spearheaded a complete overhaul of a legacy fintech system, transitioning from Laravel monolith to NestJS + Next.js microservices. Achieved 10% throughput improvement and 75% reduction in security incidents. Owned full lifecycle: UI/UX (Figma), BRD, QA (Jest, Cypress), deployment.

- Digital Wallet Stability & Compliance: Overhauled React Native mobile app, resolving critical bugs. Led technical preparation for Bank Indonesia audit, contributing to successful operational approval.
- Regulatory Integration: Actively engaged with OJK and AFPI regulators - wrote core data reporting modules ensuring 100% success rate on daily FDC synchronization.

#### **PT Maccon Generasi Mandiri** Mar 2022 - Mar 2023

**Full Stack Developer** · Makassar, Indonesia

- Cost Reduction (80%): Migrated platform from expensive managed vendor (Rp 20M/year) to self-managed VPS (Rp 4M/year) via full Laravel/PHP rebuild.
- Product Growth: Integrated Zoom API for webinars and group sessions, driving 25% increase in monthly signups within 3 months.

### **KEY PROJECTS & OPEN SOURCE**

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**AuraFlow AI** · Python · LangGraph · Node.js · BullMQ · Redis

- Distributed multi-agent backend system using LangGraph to orchestrate autonomous parse-validate workflows - replacing manual multi-step processes with a self-healing async pipeline. Node.js gateway + BullMQ queue + Python LangGraph worker.

**LedgerFlow** · NestJS · PostgreSQL · Redis · Docker

- High-performance digital wallet API with Optimistic Concurrency Control (OCC) and atomic transactions to prevent race conditions and double-spending. Two-tier JWT + Redis auth with session kill-switch.

**Portfolio Backend (awaluddin.dev)** · NestJS · Prisma · Aiven PostgreSQL · Upstash Redis · OpsCtrl

- Database-driven portfolio backend with dynamic content management. Deployed on OpsCtrl (Kubernetes), integrated with Aiven PostgreSQL via SSL CA certificate and Upstash Redis. Full CI/CD via GitHub.

**Vertical Slice Architecture Boilerplate** · Node.js · Express · Inversify · Open Source

- Production-ready starter kit implementing clean Vertical Slice Architecture for Node.js - shared openly to help teams avoid monolith pitfalls from day one.

### **EDUCATION**

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**Universitas Hasanuddin (Unhas)** 2013 - 2018

Bachelor of Engineering, Telecommunications Engineering · Makassar, Indonesia

**Xidian University** Jan - Jun 2017

Exchange Program, Information Technology · Xi'an, China